Davide Agnoletto

Contact Information

Department of Statistical Science
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Research Interests

Bayesian methodology, Bayesian predictive inference, generalized Bayes, Factor models.

Current position

Postdoctoral Associate
Duke University, Department of Statistical Science (10/2024 – present)
Advisor: David Dunson

Education

Università degli Studi di Padova, Padova, Italy

- **Ph.D. in Statistical Sciences**, Department of Statistical Sciences
  (from 10/2021 to 09/2024, thesis defense 01/2025)

  Thesis title: *Statistical modeling within the generalized Bayesian paradigm*
  Advisors: Bruno Scarpa, Tommaso Rigon, and David Dunson

- **M.Sc. in Statistical Sciences**, Department of Statistical Sciences
  (from 10/2019 to 09/2021)

  Thesis title: *Analysis of mortality curves: non-parametric density estimation using a Bayesian approach*
  Advisors: Bruno Scarpa and Tommaso Rigon
  Final mark: 110/110 with honors

- **B.Sc. in Statistics for Economics and Business**, Department of Statistical Sciences
  (from 10/2016 to 09/2019)

  Thesis title: *The use of All-Resolution Inference for fMRI images without smoothing*
  Advisor: Livio Finos
  Final mark: 110/110 with honors

Visiting Periods

Duke University, Department of Statistical Science, NC, USA (from 12/2022 to 09/2024)
Supervisor: David Dunson.

Awards

Travel & academics awards

- Travel award. 14th International Conference on Bayesian Nonparametrics, Los Angeles, USA, 2025.
- Best poster award. ISBA World Meeting, Venice, IT, 2024.
- ISBA travel award. ISBA World Meeting, Venice, IT, 2024.
- Travel award (accommodation + meal). Bayes at CIRM: Autumn school in Bayesian Statistics, Marseilles, FR, 2023.

Data competitions

- First prize winner at HackTheGene, Padova, IT, 09/2022.

Funding

- Full three-years Ph.D. scholarship (2021-2024) Department of Statistical Sciences, University of Padova.
- Mille e una Lode Award 2018/2019, scholarship awarded to the best 3% students of the University of Padova

Publications

Articles in peer-reviewed journals

- **Agnoletto, D.**, Rigon, T., and Dunson D. B. (2026). Nonparametric predictive inference for discrete data via Metropolis-adjusted Dirichlet sequences. *Journal of the American Statistical Association (Theory and Methods)*, to appear.
  https://doi.org/10.1080/01621459.2026.2662438
- **Agnoletto, D.**, Rigon, T., and Dunson D. B. (2025). Bayesian inference for generalized linear models via quasi-posteriors. *Biometrika*, 112(2).
  https://doi.org/10.1093/biomet/asaf022

Ph.D. Thesis

- **Agnoletto, D.** (2025). Statistical modeling within the generalized Bayesian paradigm. Università degli Studi di Padova.

Conference proceedings

- **Agnoletto, D.**, Rigon, T., and Dunson, D. B. (2024). Bayesian inference for generalized linear models via quasi-posteriors: an application to Eurasian chaffinch abundance in Finland. *In Methodological and Applied Statistics and Demography III. SIS 2024.* (Pollice, A., Mariani, P.) 2024, ISBN: 978-3-031-64430-6.
- **Agnoletto, D.**, Rigon, T. and Scarpa, B. (2023). Bayesian density estimation for modeling age-at-death distribution. *In Statistical Learning, Sustainability and Impact Evaluation. SIS 2023* (Chelli, F. M., Ciommi, M., Ingrassia, S., Mariani, F., Recchioni, M.C.) 2023. ISBN: 978-3-032-10629-2.

Submitted and working papers

- Carminati. A., Stolf, F., **Agnoletto, D.**, and Durante, D. (2026+). Personalized treatment selection via sequential Bayesian logistic models. In preparation.
- **Agnoletto, D.**, Rigon, T., and Dunson, D. B. (2026+). Bayesian predictive semiparametric regression. In preparation.
- **Agnoletto, D.**, Wang, P., and Dunson, D. B. (2026+). Double-covariate mean-covariance factor regression. In preparation.

Software

GitHub repositories

- GitHub repository `MAD_Sequences`: C++/R code for the paper *Nonparametric predictive inference for discrete data via Metropolis-adjusted Dirichlet sequences.*
  https://github.com/davideagno/MAD_sequences
- GitHub repository `Quasi_Posteriors`: R code for the paper *Bayesian inference for generalized linear models via quasi-posteriors.*
  https://github.com/davideagno/Quasi-posteriors

Service to profession

Referee for

- Econometrics and Statistics
- Statistics and Probability Letters

Membership to scientific societies

- International Society for Bayesian Analysis (ISBA), BNP ISBA and jISBA.
- Società Italiana di Statistica (SIS), ySIS.

Service to university

Positions in Boards and Committees

- Board member of the Department of Statistical Sciences, University of Padova (from 09/2022 to 09/2024)
  Representation of PhD students and research fellows

Mentored Students

- Peiran Wang (M.Sc., Duke University, Supervisor: David Dunson)

Teaching

Duke University, NC, USA

- Lecturer for Generative Modeling (Ph.D.) – Spring 2026 (1 guest lecture, 2 hours)

Fundings and research groups

Member of the research group

- Grant R01ES035625 entitled *Improving inferences on health effects of chemical exposures*, funded by the National Institute of Health (NIH), USA. PI: David Dunson.

Presentations

Invited presentations

- SELECT - Climbing mortality model, Misurina, IT, 2022.
  Bayesian density estimation for modeling age-at-death distribution.

Contributed presentations

- 14th International Conference on Bayesian Nonparametrics, Los Angeles, USA, 2025.
  Nonparametric predictive inference for discrete data via Metropolis-adjusted Dirichlet sequences.
- BAYSM 2024: Bayesian Young Statisticians Meeting, Venice, IT, 2024.
  Nonparametric predictive inference for discrete data via Metropolis-adjusted Dirichlet sequences.
- SIS 2024: 52nd Scientific Meeting of the Italian Statistical Society, Bari, IT, 2024.
  Bayesian inference for generalized linear models via quasi-posteriors: an application to Eurasian chaffinch abundance in Finland.
- BAYSM 2023: Bayesian Young Statisticians Meeting, online, 2023.
  Bayesian inference for generalized linear models via quasi-posteriors.
- Bayes at CIRM: Autumn school in Bayesian Statistics, Marseilles, FR, 2023.
  Bayesian inference for generalized linear models via quasi-posteriors.
- SEAS-IN 2023: Statistical Learning, Sustainability and Impact Evaluation, Ancona, IT, 2023. *Bayesian density estimation for modeling age-at-death distribution.*

Poster presentations

- ISBA World Meeting, Venice, IT, 2024.
  Bayesian inference for generalized linear models via quasi-posteriors.
- Statistical methods and models for complex data: 800 years of research to understand a complex world, Padova, IT, 2022.
  Bayesian density estimation for modeling age-at-death distribution.

Summer schools and workshops

Summer schools

- Data Research Camp, Florence, IT, 2024. Selected participant.
  Four-day meeting where groups of young scholars, advised by a senior researcher, were asked to develop innovative methods and models to analyze a common dataset.
- Bayes at CIRM: Autumn school in Bayesian Statistics, Marseilles, FR, 2023. Selected participant and presenter.

Workshops

- SELECT - Climbing mortality model, Misurina, IT, 2022.